Attention-Free Global Multiscale Fusion Network for Remote Sensing Object Detection

Tao Gao, *Member, IEEE*, Ziqi Li, Yuanbo Wen, Ting Chen, Qianqian Niu, and Zixiang Liu

***Abstract*— Remote sensing object detection (RSOD) encounters challenges in complex backgrounds and small object detection, which are interconnected and unable to address separately. To this end, we propose an attention-free global multiscale fusion network (AGMF-Net). Initially, we present a spatial bias module (SBM) to obtain long-range dependencies as a part of our proposal global information extraction module (GIEM). GIEM efficiently captures the global information, overcoming challenges posed by complex backgrounds. Moreover, we propose multitask enhanced structure (MES) and multitask feature pretreatment (MFP) to enhance the feature representation of multiscale targets, while eliminating the interference from complex backgrounds. In addition, an efficient context decoupled detector (ECDD) is presented to provide distinct features for regression and classification tasks, aiming to improve the efficiency of RSOD. Extensive experiments demonstrate that our proposed method achieves superior performance compared with the state-of-the-art detectors. Specifically, AGMF-Net obtains the mean average precision (mAP) of 73.2%, 92.03%, 95.21%, and 94.30% on detection in optical remote sensing images (DIOR), high resolution remote sensing detection (HRRSD), Northwestern Polytechnical University Very High Resolution-10 (NWPU VHR-10), and RSOD datasets, respectively.**

***Index Terms*— Attention-free, multiscale object detection, remote sensing images (RSIs), semantic information, small object detection.**

I. Introduction

OBJECT detection is a popular research task in the analysis of remote sensing images (RSIs), which is widely used in many fields, such as intelligent transportation, urban planning, military reconnaissance, and environmental monitoring [1]. Given the vast amount of remote sensing satellite data, detecting remote sensing objects in the presence of intricate backgrounds, densely clustered small objects, and extensive scale variations has been consistently a focal point of research [3]. Differing from natural scene images, RSIs are captured from a top–down perspective, leading to remote sensing objects exhibiting diverse scales and uncertain orientations, where small targets are often submerged within complex background noise.

The mainstream object detection algorithms based on deep learning can be broadly categorized into two main types: the one-stage solutions and the two-stage detectors. The two-stage algorithms first generate the proposal regions based on the input image and then perform classification. Faster-region with CNN feature (RCNN) [4] is one of the typical solutions. Despite their excellent accuracy, their performance is restricted by an enormous amount of calculation and time overhead. The one-stage algorithms regard detection as a regression problem for a single network. The typical methods include single shot multibox detector (SSD) [5], you only look once (YOLO) [6], [7], [8], [9], [10], [11], and RetinaNet [12]. Recently, the feature pyramid network (FPN) and several enhanced algorithms [13], [14], [15], [16], [17] build upon it have introduced innovative methods for integrating information from multiple feature levels. These advancements have significantly improved the accuracy of detecting targets at various scales. Moreover, some anchor-free algorithms have been proposed in recent years, such as CenterNet [18], CornerNet [19], fully convolutional one-stage object detection (FCOS) [20], and you only look once X (YOLOX) [21]. They transform object detection into key points or center point detection, without anchor parameters or proposals. It is worth noting that YOLOX proposes a decoupled head which greatly enhances convergence speed and significantly improves performance by efficiently leveraging semantic and positional information and introducing decoupled head to the YOLO family.

Differing from mainstream object detection tasks, as illustrated in Fig. 1, remote sensing object detection (RSOD) is characterized by its extensive coverage, complex environments, substantial variations in target scales, and the presence of various types of noise. These factors prone to low detection accuracy, missed detections, and false positives. Given the differences between generic objects and remote sensing objects, many researchers have worked on multiscale object detection and feature fusion based on FPN, targeting directly the challenge of significant scale variations in RSOD. Kang et al. [22] and Liu et al. [23] perform RSOD based on the RPN method. Due to the slower inference speeds of two-stage network, many researchers have adopted one-stage network for RSOD. Wu et al. [24] devise a superior "U-Net in U-Net" framework, treating detection as a segmentation problem and employing multiscale learning to achieve excellent multiscale object detection in remote sensing without interference. However, this architecture is not conducive to efficiently processing a large volume of RSIs. Wang et al. [25], Ma et al. [26], Wu et al. [27], Yi et al. [28], and Mei et al. [29] have utilized the concept of multiscale feature enhancement in FPN to highlight target features at various scales. However, the aforementioned work focuses solely on feature enhancement, overlooking the

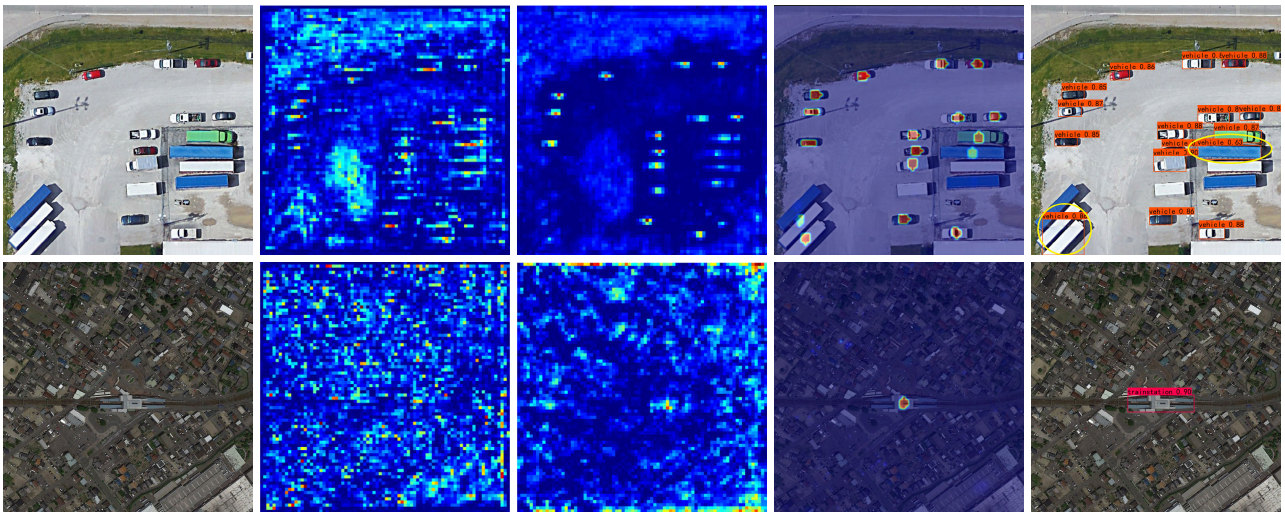
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The authors are with the School of Information Engineering, Chang'an University, Xi'an 710064, China (e-mail: gtnwpu@126.com; lzq@chd.edu.com; wyb@chd.edu.cn; tchenchd@126.com; 1273200630@qq.com; liuzixiang_ah@163.com).

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(a) (b) (c) (d) (e)

Fig. 1. Visualizing the RSI object detection. RSOD is marked by its wide coverage range, leading to complex environmental interferences, such as geographic and architectural textures, variations in light and shadow, cloud obscuration, and the diverse shapes of multiscale objects, including shape distortion and extreme aspect ratios. (a) Two original RSIs. (b) Characteristics of complex background noise in the heatmap. (c) Features after enhancement and denoising. (d) Distribution of features on the original image. (e) Final detection results. Yellow circles denote the false detection.

denoising aspect in the context of RSIs. To the end, a brand new neck network multitask enhanced structure (MES) has been proposed to integrate the enhanced semantic information into multiscale feature maps and filter noise in complex backgrounds. The experimental results demonstrate that MES effectively filters out a significant amount of noise while highlighting the target features. It utilizes multiple multitask feature pretreatment (MFP) modules and the feature enhancement techniques to improve the representation of target information, while employing the advanced fusion strategies to output the multiscale feature maps for detectors.

Despite progress, they overlook the global semantic information, which plays a crucial role in understanding the background, objects, and their relationships within the RSIs [30], [31], [32]. Due to the size of the limited receptive field, convolutional neural networks suffer from learning long-range dependencies. Subsequently, some attention mechanisms have been introduced into RSOD to improve the detection performance. Liu et al. [33] preserve the relationships of the entire dataset and capture the dependencies between the space and channel. However, learning the dependencies between datasets inevitably introduces additional computational overhead. Subsequently, Ma et al. [34], Wu et al. [35], Gao et al. [36], Hu et al. [37], Zhang et al. [38], Wen et al. [39], and Mei et al. [40] introduce attention mechanisms into RSOD to guarantee a better understanding of global knowledge. However, self-attention operations still suffer from a fatal flaw that their computational cost is quadratic $O(n^2)$, leading to a slowdown in inference speed.

Therefore, to tackle challenges including insufficient global information extraction from RSIs, limited feature representation, and high computational demands of attention mechanisms, we introduce a novel global information extraction module (GIEM), comprising several spatial bias modules (SBM) that acquire global knowledge without relying on self-attention. Different from the existing methods using the heavy self-attention module, our proposed method learns the global knowledge by adding a few spatial bias channels to the convolution feature map.

Moreover, many researchers focus on designing detection heads to improve the accuracy of detecting small objects. For instance, Wang et al. [41] propose a multiple center point-based learning network (M-CenterNet) to improve the localization performance of tiny object detection. Disentangle your dense object detector (DDOD) [42] employs deformable convolutions to dynamically select specific spatial features for each head, emphasizing the importance of separating classification and localization tasks. However, most of the aforementioned methods [43], [44], [45] utilize coupled or anchor-based detection heads, significantly impacting detection efficiency and accuracy [21]. Subsequently, decoupled detection heads, especially anchor-free ones, have gradually become the mainstream for object detection tasks. Nevertheless, the aforementioned researches do not fully exploit contextual information which is revealed by our experiments. Therefore, developing a detection head specialized for RSOD tasks to extract objects from complex backgrounds holds promising prospects.

For these reasons, we present an anchor-free efficient context decoupled detector (ECDD) specifically designed for RSIs. ECDD enhances the encoding features for both classification and localization tasks. We introduce SBM into ECDD to obtain context information. The decoupled structure of ECDD provides a high-resolution feature map containing more edge information to better regress object boundaries for localization. Furthermore, ECDD provides features with rich semantic context for classification tasks, facilitating the inference of object categories. Finally, we optimize loss functions to enhance detection performance.

We summarize our main contributions as follows.

1) With our proposed SBM, we develop a GIEM, which enhances the representation of multiscale targets.

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2) We design an MES, aiming to inject salient information to different feature levels for further enhancing the representation of multiscale targets, while eliminating the interference from complex backgrounds.
3) We propose an ECDD tailored for RSIs, which provides the fused semantically strong and high-resolution features for classification and localization tasks separately. Moreover, we optimize the loss functions to emphasize the positive samples.

II. Related Works

A. Global Semantic Information Exploitation

The exploration of contextual information is necessary to comprehend the relationship between complex environments and goals. Many ground-based methods proved that exploiting the contextual relationship is beneficial to work out the problems of insufficient representation of object appearance and alleviate environmental interference. Therefore, many effective modules are introduced to extract local and global contextual information. FPN [13] utilizes a top–down pathway to transmit semantic information from the deep feature map to enhance the shallower feature maps across various scales which shows the significance of utilizing semantic information for object detection. Tian et al. [31], Fu et al. [46], Gao et al. [47], and Cao et al. [48] also show the importance of using semantic information for object detection. Therefore, it is more crucial to mine and extract semantic information for RSIs with complex backgrounds. Dai et al. [49] design a feature fusion method by combining the bottom-to-up and up-to-bottom ways to strengthen the positioning and semantic feature of multiscale ships, particularly small ships [36]. Spatial attention module (SAM) [50] is proposed to exploit local contextual expression, while transformer [51] was created to capture global relations between pixels. Wu et al. [24] propose an interactive-cross attention module that encodes local contextual information between low-level details and high-level semantic features through the outputs of two subtasks, cross channel attention, and interactive-cross spatial attention. Fang et al. [52] propose a location-based attention method that distills positive information in a global context. Especially for RSIs, lighting, shadows, and similar objects in the overhead imaging process collectively influence the detection efficiency. Full-scale object detection network (FSoD-Net) [25] establishes three branches to capture various contextual information for multiscale object detection.

However, these aforementioned methods extract and enhance the semantic information by heavy design of attention, conventional nonlocal methods are inserted only at specific layers in a convolutional neural network by consideration of the throughput and network size. In addition, since using attention mechanisms to acquire nonlocal information operations only considers spatial correlation by merging channels, they are blind to any channel correlation. Therefore, to overcome these limitations, we design the SBM as a part of our proposal GIEM, enabling the network to enhance global knowledge exploitation and strengthen the semantic information contained in the deep feature map.

B. Multiscale Information Fusion

FPN as mentioned earlier innovatively incorporates and outputs information at different scales through a top–down path, which greatly improves the detection performance of the network for multiscale targets. Since then, many improved structures based on FPN have gradually appeared. For example, path aggregation network (PANet) [14], bidirectional FPN (BiFPN) [53], and one-shot PANet architecture search (OPANAS) construct various heterogeneous information paths to aggregate multiscale information efficiently. Recently, many FPN-based improved algorithms have achieved remarkable achievement in view of the characteristics of RSIs. For example, Liu et al. [23] propose scene-enhanced FPN (SE-FPN) to capture contextual features of objects with the help of the scene features. Sun et al. [54] propose a novel bidirectional feature fusion module (Bi-DFFM) that can effectively aggregate multiscale features by bidirectional (top–down and bottom–up) information interaction, which is helpful for detecting ships of various scales. Kang et al. [22] design a contextual region-based CNN with multilayer fusion, which fuses the deep semantic and shallow high-resolution features by using low-resolution feature maps from a single layer for proposal generation in a region proposal network (RPN). In order to combine multiscale features adaptively in different channels and spatial positions, Liu et al. [30] introduce adaptive FPN (AFPN) to obtain more discriminative features. Nah et al. [55] present multiscale loss function that mimics conventional coarse-to-fine approaches to enhances convergence greatly. Zhang et al. [56] leverage the semantic and geometric information as priors to train the multiscale networks. Wang et al. [57] propose a feature-reflowing pyramid (FRP) structure to generate high-quality feature presentation for each scale by fusing fine-grained features from the adjacent lower level.

In this article, we propose that MES to generate high-quality feature presentations for each scale by fusing fine-grained features from the adjacent lower level. This structure incorporates MFP and GIEM, which effectively extinguish noises and highlight targets in RSIs.

C. Small Object Detection in RSIs

Small object detection has always been challenging in the field of object detection for two main reasons. Initially, a small object typically occupies only a fraction of the entire image, often spanning just a few pixels or even less. Moreover, the features of small objects may become overwhelmed as the network depth increases. Subsequently, RSIs come with extensive and intricate backgrounds, which introduce undesirable yet unavoidable interference when it comes to detecting small objects. Therefore, researchers have carried out extensive studies on small object detection. Deng et al. [58] propose the extended FPN (EFPN) with an extra high-resolution pyramid level specialized for small object detection. Chen et al. [59] designed a parallel bifusion feature pyramid structure with a bottom–up fusion module (BFM) to detect both small and large objects at once with high accuracy. Yang et al. [60] present a sampling fusion network (SF-Net), which fuses multilayer features with effective anchor sampling to improve

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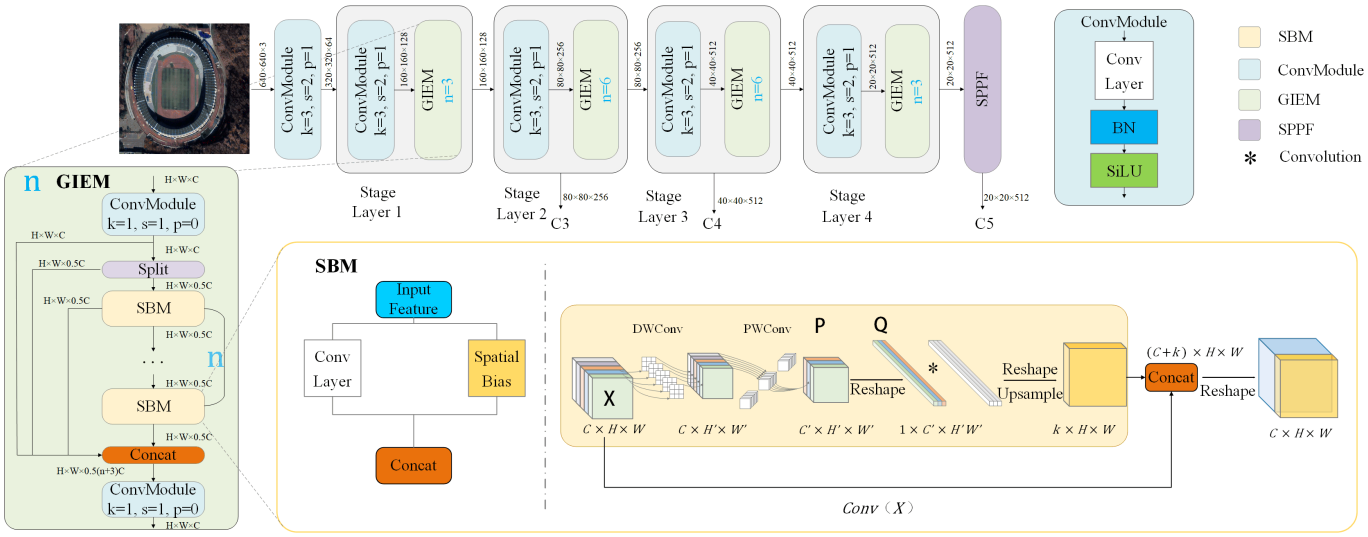


Fig. 2. Overall architecture of our backbone. It has four hierarchical stages, each with a stack of GIEM blocks and preceded by a ConvModule. The last three layers are used for feature classification. Within each GIEM block, contains two ConvModules and n SBMs. Every ConvModule is composed of a Conv2d layer followed by batch normalization and SiLU activation layer. The figure on the left of SBM shows the overall workflow, and the right shows its detail. To capture the global dependency, the channel and spatial size of the feature map are reduced through the 1×1 convolution and average pooling operation. We aggregate spatial bias on a reduced feature map using a simple 1-D convolution operation.

the sensitivity of small objects. The aforementioned work effectively leverages the FPN structure to separate semantic information and positional information. It not only fuses detail-preserving low-level features and semantic-rich high-level features but also distributes the classification and localization tasks of different objects to corresponding feature maps according to their scale on images. While the insightful point [27] that the feature maps in low layers strongly respond to edges or instance parts manifests the necessity of augmenting a bottom–up path to propagate features with strong boundary information and enhance all features with reasonable localization capability. YOLOX [21] introduces a decoupled head that significantly enhances convergence speed and improves performance by efficiently leveraging semantic and positional information. Building upon the concept of decoupled heads, DDOD [42] suggests the utilization of deformable convolutions to train separate convolutional offsets for each branch. This approach aims to dynamically select distinct spatial features for each head. These studies underscore the significance of decoupling classification and localization tasks. However, a satisfactory unified solution that combines the FPN structure to address the requirements for both semantic features and high-resolution features in decoupled detection heads for classification and localization tasks has not been provided.

In this article, ECDD is designed specifically for RSIs which fully utilize the combined semantically robust features and high-resolution feature map provided by the MES structure to enhance both classification and localization tasks to improve the accuracy of small object detection.

III. Methodology

A. GIEM

To attain a more comprehensive understanding of environmental relationships and detect targets in RSIs, especially in vast geographical areas with intricate environmental factors, we propose GIEM as a feature extraction module to better capture global contextual information. As illustrated in Fig. 2, we indicate an SBM to obtain global long-range information. In order to create the spatial bias, we incorporate long-range dependencies by reducing the dimensionality of feature maps in both channel and spatial directions. Subsequently, SBM extends this approach by combining the spatial bias map with the convolutional feature map in the channel dimension. This enables the network to integrate global knowledge from the spatial bias with the local features from the convolutional feature map, allowing it to learn both local and global information simultaneously. The spatial bias captures a broader region, while the convolutional features focus on local details within an image. Our feature extraction backbone network is composed of four GIEM modules, which are capable of extracting global semantic information from different depths. SBM is a crucial component of the GIEM residual connections, where the blue-lettered "n" represents the number of SBMs used in different depths of GIEM modules.

Consider the input feature map X of a convolutional layer denoted by $X \in \mathbb{R}^{C\times H\times W}$. We compress the input feature map in both the channel and spatial dimensions. Specifically, we utilize a Depthwise convolution (DWConv) [DWDWDW] to transform the size of the feature map to $C \times H' \times W'$. Then, we adopt a pointwise convolution (PWConv) layer for the spatial compression that generates $P \in \mathbb{R}^{C'\times H'\times W'}$. We obtain a transformed feature map by flattening each channel of the feature map P into a 1-D vector $Q \in \mathbb{R}^{1\times C'\times H'W'}$

$$P = \text{PWConv}(\text{DWConv}(X)) \tag{1}$$

$$Q = F_{\text{Reshape}}(P) \tag{2}$$

where X denotes the input feature map, DWConv$(\cdot)$ and PWConv$(\cdot)$ denotes DWConv and PWConv, and $F_{\text{Reshape}}(\cdot)$ denotes flattening operation.

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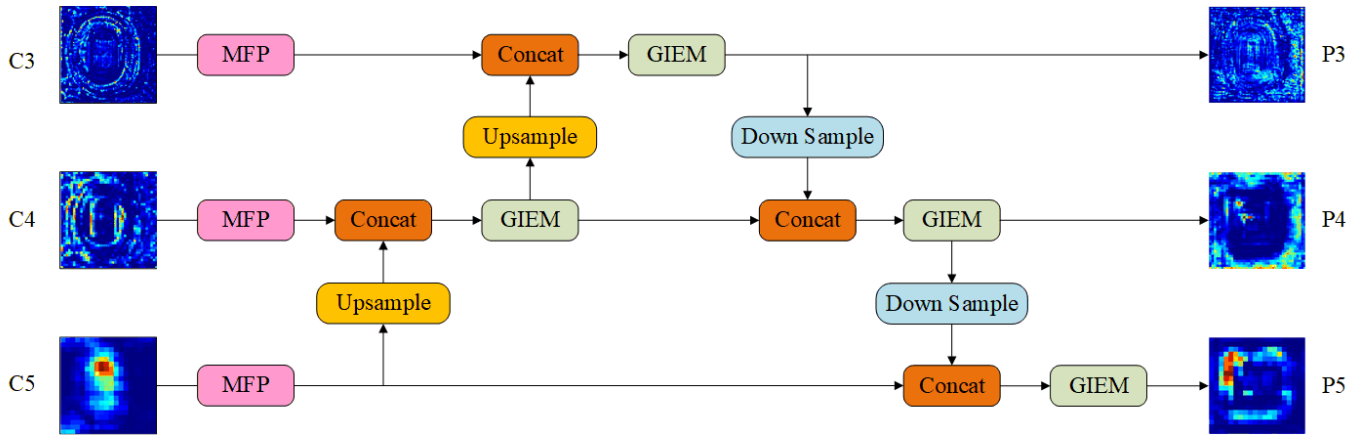


Fig. 3. Overall architecture of MES.

Subsequently, to gather global knowledge from this transformed feature map, we utilize a $1 \times N$ convolution in the channel dimension to encode the interchannel relationships of global knowledge

$$\mathrm{SB} = F_{\mathrm{bilinear}}(\mathrm{Conv}_{1d}(Q, N)) \tag{3}$$

where $\mathrm{Conv}_{1d}(\cdot, N)$ denotes a $1 \times N$ convolution and $F_{\mathrm{bilinear}}(\cdot)$ denotes the bilinear interpolation.

Moreover, we utilize bilinear interpolation to perform upsampling on the spatial bias feature map, aligning it with the size of the convolutional feature map, after which we concatenate these two feature maps

$$\mathrm{SBM}(X) = \mathrm{ReLU}(\mathrm{BN}[\mathrm{Conv}(X), \mathrm{SB}]) \tag{4}$$

where $\mathrm{Conv}(\cdot)$ and SB denote a 3×3 convolution and a spatial bias, $[\cdot,\cdot]$ represents the concatenation, and $\mathrm{SBM}(\cdot)$ denotes the output of SBM. Finally, the resulting feature map is reshaped to attain a size of $C \times H \times W$.

We construct the core components of GIEM by assembling two ConvModules, each comprising standard convolution, batch normalization, and sigmoid linear unit (SiLU) activation layers, along with a SBM. The output of the ConvModule is represented as follows:

$$\mathrm{ConvM}(X) = \mathrm{SiLU}(\mathrm{BN}[\mathrm{Conv}(X)]) \tag{5}$$

where $\mathrm{ConvM}(\cdot)$ denotes the ConvModule.

In the GIEM, the input feature map $X \in \mathbb{R}^{C \times H \times W}$ undergoes a ConvModule, resulting in the output $X \in \mathbb{R}^{C \times H \times W}$. Then, the output is split into two parts which halve the number of channels, with one feature map serving as input to the SBM, and the other as input for feature fusion. Subsequently, the input feature map $X \in \mathbb{R}^{0.5C \times H \times W}$ is sequentially passed through n SBMs, retaining the output features from each SBM. Afterward, these output features are concatenated together to obtain $X \in \mathbb{R}^{0.5(n+3)C \times H \times W}$. Finally, we pass the fused feature map through a ConvModule to resize the output feature map to $C \times H \times W$

$$\mathrm{GIEM}(X) = \mathrm{ConvM}\big[\mathrm{ConvM}(X), \mathrm{Sp}(X), \mathrm{SBM}(\mathrm{Sp}(X)), \ldots, \mathrm{SBM}(\cdots \mathrm{SBM}(\mathrm{Sp}(X)))\big] \tag{6}$$

where $\mathrm{Sp}(\cdot)$ denotes the split operation which splits the input into two along the channel dimension, and $\cdots$ encompasses the operations of passing through n SBMs, which have been omitted for brevity. Each branch is concatenated to obtain the output feature map $X \in \mathbb{R}^{0.5(n+3)C \times H \times W}$, which is subsequently processed to produce the output $X \in \mathbb{R}^{C \times H \times W}$ through ConvModule.

We concatenate ConvModules and n GIEMs at each stage as depicted in Fig. 2. Different stages employ varying quantities of GIEMs. The outputs from stages 3 and 4 are used as inputs C3 and C4 for the subsequent feature fusion operation within the MES structure. In addition, the output from stage 4 undergoes pooling operation spatial pyramid pooling-fast (SPPF) [61] to yield C5.

B. MES

1) Multiscale Feature Fusion: RSIs, with their intricate environments and small targets, often exhibit low-resolution and limited discriminative feature information. Moreover, they are highly susceptible to interference from environmental factors. Achieving adequate and balanced feature representation is thus essential for effective multiscale object detection in RSIs. As illustrated in Fig. 3, MES includes MFP which significantly reduces the noise for the features and enhances feature representation. GIEM plays an indispensable role in learning global knowledge and improving information exchange between deep and shallow layers. This optimization in information exchange within the network architecture can provide fused semantically strong features and high-resolution feature maps for classification and localization tasks, thereby enhancing the network's detection efficiency. Subsequent experiments have demonstrated that MES effectively enhances the detection accuracy of RSOD.

2) Feature Pretreatment: Initially, we introduce MFP as the primary tool for feature enhancement and noise reduction to address the challenge of poor representation capability of remote sensing targets in complex backgrounds. Unlike prior works, the MFP module achieves both a global receptive field and multiscale learning with efficient operations. Furthermore, MFP could denoise input features for the MES.

We aggregate information from three paths (Q, K, V) to obtain multiscale feature maps. The aggregation process is illustrated in Fig. 4. This information aggregation process is independent for each Q, K, and V in each path. PConv for

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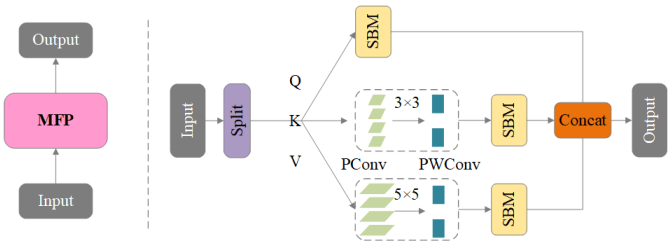


Fig. 4. Overall architecture of MFP. (Left) Building block of MFP consists of several SBMs and partial convolution (PConv) modules. The MFP module focuses on capturing context information and specializing in reducing background noise. (Right) After obtaining Q, K, and V features through linear projection, we create multiscale features by combining nearby feature information using PConv. These multiscale features are processed with SBM and then concatenated before being passed through the final linear projection layer for feature fusion.

information aggregation to avoid hurting hardware efficiency

$$Q/K/V = \mathrm{Sp}(X) \tag{7}$$

$$\mathrm{MFP}(X) = \big[\mathrm{SBM}(Q), \mathrm{SBM}(\mathrm{PWConv}(\mathrm{PConv}_{3\times3}(K)), \mathrm{SBM}(\mathrm{PWConv}(\mathrm{PConv}_{5\times5}(V))\big] \tag{8}$$

where $\mathrm{MFP}(\cdot)$ denotes the MFP module, $\mathrm{PConv}_{n\times n}(\cdot)$ denotes the PConv, and $n \times n$ denotes the size of convolutional kernels. After that, we utilize SBM to perform global attention upon them to extract multiscale global features. Finally, we concatenate the features from different scales to fuse the features.

Classification and localization are two highly related but still contradictory tasks in object detection [21], [43], [62], [63]. Classification involves a higher level semantic analysis, while localization demands a finer-grained assessment, emphasizing the importance of precise boundary information. Specifically, given a ground-truth object assigned to a specific pyramid level l, with bounding box B and class label c, mainstream detectors utilize a decoupled head to minimize the classification and localization loss based on the same pyramid-level feature map P_l

$$L = L_{\mathrm{cls}}(F_c(P_l, c) + L_{\mathrm{loc}}(F_r(P_l, B) \tag{9}$$

where L denotes the loss, $F_c(\cdot) = \{f_{\mathrm{cls}}(\cdot), C(\cdot)\}$, $F_r(\cdot) = \{f_{\mathrm{loc}}(\cdot), R(\cdot)\}$ are the classification and localization branches. $f_{\mathrm{cls}}(\cdot)$ and $f_{\mathrm{loc}}(\cdot)$ are the feature projection functions for classification and localization, while $C(\cdot)$ and $R(\cdot)$ are the final layers in two branches which decode features to classification scores and bounding box positions. In the common decoupled head design, $f_{\mathrm{cls}}(\cdot)$ and $f_{\mathrm{loc}}(\cdot)$ share an identical structure but are trained with separate parameters to impart distinct feature contexts to each task, a process known as parameter decoupling. We found that introducing SBM, which can increase global feature extraction, into the detection head can effectively improve detection. Therefore, our network introduces an ECDD tailored for RSIs which utilizes two SBMs and two 1×1 Convs to generate SOTA feature maps for classification and localization tasks.

C. Efficient Context Decoupled Head

To address the RSOD task, we propose ECDD, which enhances detection accuracy by utilizing high-resolution and strong semantic feature maps. In addition, we have further optimized the loss function for ECDD. As illustrated in Fig. 5, our method follows the common one-stage detection framework, receives three levels of feature maps, and generates decoupled feature maps for classification and localization. For the loss function, focal loss [12] is often used to resolve the problem of severe imbalance between positive and negative samples and prevents the vast number of easy negatives from overwhelming the detector during training

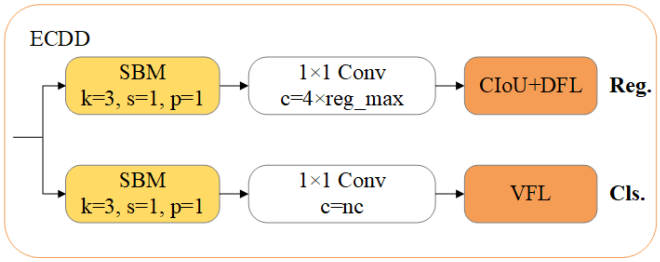


Fig. 5. Overall architecture of ECDD. It decouples the input prediction feature map into two branches by SBM and 1×1 convolution layer. The two branches are used for classification and regression tasks.

$$L_{\mathrm{FL}}(p, y) = \begin{cases} -\alpha(1-p)^{\gamma}\log(p) & \text{if } y = 1 \\ -(1-\alpha)p^{\gamma}\log(1-p) & \text{otherwise} \end{cases} \tag{10}$$

where $y \in \pm 1$ specifies the ground-truth class and $p \in [0, 1]$ is the predicted probability for the foreground class. However, focal loss introduces hyperparameters for both positive and negative samples, which may result in a decrease in the quality of positive samples. Therefore, we introduce the variable focal loss [64] to achieve asymmetric weighting of positive and negative samples, emphasizing positive samples as the main focus

$$L_{\mathrm{VFL}}(p, q) = \begin{cases} -q(q(\log(p)) + (1-q)\log(1-p)) & q > 0 \\ -\alpha p^{\gamma}\log(1-p) & q = 0 \end{cases} \tag{11}$$

where $L_{\mathrm{VFL}}()$ denotes variable focal loss, p is the predicted intersection over union (IoU)-aware classification score, q is the target score, and α and γ are the weight coefficient. The varifocal loss only reduces the loss contribution from negative examples ($q = 0$) by scaling their losses with a factor of p^{γ}. When $q > 0$ in VFL, there are no hyperparameters for positive samples, meaning that there is no attenuation. However, when $q = 0$, VFL introduces hyperparameters for negative samples. γ reduces the contribution of negative samples, and α prevents excessive suppression. Overall, this results in a reduction of the negative samples' contribution.

In addition, taking into account that the aspect ratio of the bounding boxes can affect detection performance, we utilize the complete intersection over union (CIOU) loss function as follows:

$$L_{\mathrm{CIoU}} = \frac{1 - \mathrm{IoU} + \rho^2\left(b.b^{gt}\right)}{c^2} + \alpha v \tag{12}$$

where IoU measures the overlap between two rectangles. b and b_{gt} are the center points, ρ is the euclidean distance, c is the diagonal distance of the closed area, v measures relative proportions, and α is a weighting coefficient.

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Finally, to quickly focus the network on positions closer to the target, we introduce the distance focal loss (DFL) [65] function

$$L_{\mathrm{DFL}}(P_i, P_{i+1}) = -((y_{i+1} - y)\log(P_i) + (y - y_i)\log(P_{i+1})) \tag{13}$$

where $L_{\mathrm{DFL}}()$ denotes DFL. DFL is designed to amplify the probabilities of values in the vicinity of the target y(i.e., y_i and y_{i+1}). The global minimum solution of DFL, i.e., $P_i = (y_{i+1} - y/y_{i+1} - y_i)$ and $P_{i+1} = (y - y_i/y_{i+1} - y_i)$, can guarantee the estimated regression target $\hat{y}$ infinitely close to the corresponding label y which also ensures its correctness as a loss function. The indication of $\hat{y}$ is as follows:

$$\hat{y} = \sum_{j=0}^{n} P(y_i)y_i = \frac{y_{i+1} - y}{y_{i+1} - y_i}y_i + \frac{y - y_i}{y_{i+1} - y_i}y_{i+1}. \tag{14}$$

Therefore, our loss function is utilized in the following manner:

$$L_{\mathrm{cls}} = L_{\mathrm{VFL}} \tag{15}$$

$$L_{\mathrm{loc}} = L_{\mathrm{CIoU}} + L_{\mathrm{DFL}} \tag{16}$$

$$L = L_{\mathrm{cls}}(F_c(P_l, c) + \lambda L_{\mathrm{loc}}(F_r(P_l, B) \tag{17}$$

$$P_{\mathrm{reg}} = \mathrm{Conv}_{1\times1}(\mathrm{SBM}(P_l))$$

$$P_{\mathrm{cls}} = \mathrm{Conv}_{1\times1}(\mathrm{SBM}(P_l)) \tag{18}$$

where $\mathrm{Conv}_{1\times1}(\cdot)$ denotes a 1×1 convolutional layer. P_l as the common input, we feed the two task-specific branches with input features. Then, we obtain feature P_{cls} and P_{reg} for classification and regression.

IV. Experiments

A. Implement Specifications

We utilize the mosaic data augmentation technique with a probability of random mosaic set to 0.5. A stochastic gradient descent (SGD) optimizer with a momentum of 0.9 is employed. The learning rate starts at 0.001 and decays with cosine annealing to 0.0001. The training batch size is 6, and inputs from four datasets are resized to 640 × 640 pixels. Our experiments are conducted in the PyTorch framework on a Windows operating system, utilizing a workstation equipped with an RTX4070Ti 12-GB GPU and 32-GB RAM.

B. Datasets

1) Detection in optical remote sensing images (DIOR) Dataset [85]: It is the largest large-scale benchmark dataset currently utilized for object detection in optical RSIs. Targets have scale variations, extreme aspect ratios, high levels of interclass similarities, and intraclass diversity. Furthermore, the environment in the dataset is complicated, including photograhs of various sceneries and weather conditions, which present huge challenges for evaluating the performance of detection models. The size of images in this dataset is 800 × 800, with a spatial resolution range of 0.5–30 m. There are a total of 20 ground target categories in the dataset, and they are, respectively, airplane (AL), airport (AT), baseball field (BF), basketball court (BC), bridge (B), chimney (C), dam (D), expressway service area (ESA), expressway toll station (ETS), harbor (HB), golf course (GC), ground track field (GTF), overpass (O), ship (S), stadium (SD), storage tank (ST), tennis court (TC), train station (TS), vehicle (V), and windmill (WM). According to the instructions of the original paper, the entire dataset is divided into 5862 training images, 5863 validation photographs, and 11 738 test images.

2) High resolution remote sensing detection (HRRSD) Dataset [86]: It is a large-scale remote sensing dataset containing 21 761 satellite images from Google Earth and Baidu Map, with a spatial resolution ranging from 0.15 to 12 m. There are 55 740 instances of targets in HRRSD, whose highlight is that the sample size is relatively balanced among the categories, with about 4000 samples in each category. HRRSD is composed of 13 classes, and their species and proportions are as follows: ship (8.90%), bridge (8.20%), GTF (6.70%), ST (7.90%), BC (6.95%), TC (7.71%), AL (8.79%), baseball diamond (7.24%), HB (6.70%), vehicle (8.53%), crossroad (7.22%), T junction (7.93%), and parking lot (8.90%). To confirm that our experimental results are more representative, HRRSD is divided into three parts, with 5401 images as a training set, 5417 images as a validation set, and 10 943 images as a test set, and each of the subsets maintains a relatively balanced number of images and annotated object instances.

3) Northwestern Polytechnical University Very High Resolution-10 (NWPU VHR-10) Dataset [87]: It is an object detection dataset released by Northwestern Polytechnical University and is currently the most used in the field of remote sensing. In this dataset, 715 images with a spatial resolution of 0.5-2 m are captured from Google maps, and the remaining 85 images with a spatial resolution of 0.08 m are acquired from sharpened infrared images. And, the image sizes of this dataset are ranging from 533 × 597 to 1720 × 1028 pixels. This dataset consists of 150 negative samples and 650 samples with ten species. And, their categories and proportions are as follows: AL (19.43%), ship (7.75%), ST (16.81%, ST), baseball diamond (10.01%, BD), TC (13.45%, TC), BC (4.08%, BC), GTF (4.18%, GTF), HB (5.57%), bridge(3.18%), and vehicle (15.35%). To demonstrate the effectiveness of our proposed method, we divide the dataset into training and test sets in a ratio of 0.75:0.25.

4) RSOD Dataset [79]: It is a dataset for object detection in RSI analysis published by Wuhan University in 2015. It contains a total of 2326 images, as well as four categories: ALs, playgrounds, overpasses, and oil tanks. Their distribution in the dataset is as follows: 446 AL images containing 4993 instances, 189 playground images containing 191 playgrounds, 176 overpass icons containing 180 overpasses, and 165 oil tank images containing 1586 oil tanks. The image size in the dataset varies from 512 × 512 to 1961 ×1193 pixels. In order to illustrate the effectiveness of our proposed method, we divide it into training sets and test sets according to the ratio of 0.6:0.4.

C. Ablation Study and Analysis

We perform four ablation experiments on DIOR, HRRSD, NWPU VHR-10, and RSOD to prove the effectiveness of GIEM, MES, and ECDD. Tables I–IV show the results of four ablation experiments.

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TABLE I
ABLATION EXPERIMENT ON DIOR

Method	mAP	AL	AT	BF	BC	B	C	D	ESA	ETS	GC	GTF	HB	O	S	SD	ST	TC	TS	V	WM
Baseline	70.2	88.8	67.4	79.2	88.7	39.0	79.0	52.0	62.4	58.3	67.6	75.7	64.7	59.4	91.1	77.3	73.3	90.0	53.3	53.8	82.3
+GIEM	72.4	90.5	73.7	78.8	88.4	42.8	81.3	57.4	67.3	62.0	71.2	89.6	64.2	61.1	91.5	73.9	76.7	89.9	59.7	57.0	82.4
+MES	72.1	91.1	72.7	78.9	89.1	43.4	80.8	56.4	66.9	60.9	70.2	77.1	64.9	61.2	91.5	69.0	75.8	90.0	60.9	58.6	82.8
+GIEM+MES	72.6	90.8	72.4	78.2	88.8	42.7	81.6	58.3	67.7	63.6	70.9	78.1	64.2	60.6	91.2	74.8	76.3	90.3	61.5	57.4	82.3
+GIEM+MES+ECDD	73.2	90.9	72.8	79.3	89.7	44.7	81.4	59.3	66.0	62.7	73.8	79.2	65.0	61.7	91.7	78.6	75.8	90.7	60.0	58.0	83.1

TABLE II
ABLATION EXPERIMENT ON HRRSD

Method	mAP	AL	S	ST	BD	TC	BC	GTF	HB	B	V	CR	TJ	PK
Baseline	90.49	99.79	92.76	97.84	89.85	94.99	79.70	98.74	93.42	91.99	95.52	92.72	80.51	68.58
+GIEM	91.24	99.71	94.12	97.92	91.44	94.66	79.88	99.00	94.20	93.48	96.35	93.35	80.78	71.38
+MES	91.76	99.74	94.25	97.84	92.18	95.26	81.70	99.11	94.31	94.01	96.58	93.69	82.25	72.00
+GIEM+MES	91.44	99.74	93.70	98.32	92.60	94.74	79.58	98.96	95.13	93.13	95.67	93.58	82.41	71.13
+GIEM+MES+ECDD	92.03	99.79	93.41	98.05	92.66	96.12	82.79	99.06	93.99	93.82	96.60	94.09	84.02	71.93

TABLE III
ABLATION EXPERIMENT ON NWPU VHR-10 DATASET

Method	mAP	airplane	ship	ST	BD	TC	BC	GTF	harbor	bridge	vehicle
Baseline	91.81	99.86	94.96	96.13	97.16	98.79	87.07	99.39	92.46	75.09	77.15
+GIEM	94.36	99.54	93.96	93.40	98.55	99.58	96.33	99.00	92.38	84.31	86.52
+MES	93.02	99.73	94.71	93.65	97.91	98.74	93.17	98.11	91.77	79.37	83.02
+GIEM+MES	94.76	99.84	95.31	95.06	98.16	97.45	98.17	99.38	95.32	80.92	88.02
+GIEM+MES+ECDD	95.21	99.88	94.23	95.47	98.43	99.69	96.34	99.33	96.88	84.57	87.25

TABLE IV
ABLATION EXPERIMENT ON RSOD

Method	mAP	Aircraft	Oil tank	Overpass	Playground
Baseline	93.50	96.07	98.68	79.99	99.26
+GIEM	93.97	95.74	98.54	82.60	99.01
+MES	94.02	96.07	98.70	81.94	99.37
+GIEM+MES	94.18	96.36	99.05	82.82	98.53
+GIEM+MES+ECDD	94.30	96.02	99.02	82.43	99.7

1) Baseline: Initially, we use DarkNet53 to extract multiscale feature maps of RSIs. The FPN is applied to fuse features. Conventional decoupling head is utilized to detect targets. In all experiments, all hyperparameters are consistent for a fair comparison.

Finally, the baseline achieves 66.9% mAP on DIOR, 90.49% mAP on HRRSD, 91.81% mAP on NWPU VHR-10, and 93.53% mAP on RSOD.

2) Effect of GIEM: The proposed GIEM is designed to enhance high-level semantic information in the deep feature map, improving the network's ability to distinguish between background and objects. Our model, when equipped with GIEM, achieves a 72.4% mAP (2.2% ↑) as shown in Table I on DIOR dataset. Every category experiences an increase in its corresponding AP, with a notable 6.3% improvement in the aircraft category and a substantial 5.4% improvement in the dam category. This highlights GIEM's effectiveness in enhancing the network's comprehension of global image information, resulting in clearer object recognition and improved detection accuracy for large-scale objects, such as overpasses, as it operates on the deep feature map.

2) Effect of MES: MES is designed to enhance object feature extraction by fusing multiscale features and filtering out background interference. When replacing FPN in the baseline network, MES leads to a performance improvement of 1.27% and 1.21% mAP on the HRRSD and NWPU datasets, respectively. Furthermore, in HRRSD, MES enhances the feature representation of small objects in the feature maps, leading to a notable 1.49% improvement in the AP of the ship category. In addition, APs for baseball diamond and bridge are improved by 2.33% and 2.02%, respectively, compared with the baseline. Combining both GIEM and MES, our method achieves significant improvements in NWPU, with a 2.95% increase in mAP and 0.35%, 1.0%, 11.1%, and 5.83% higher APs for ship, baseball diamond, BC, and bridge, respectively, compared with the baseline. This showcases the effectiveness of GIEM and MES in multiscale RSIs, enabling our approach to achieve outstanding detection results.

3) Effect of ECDD: Regarding the ECDD detection head, its original purpose is to address the misalignment issue between classification and positioning tasks caused by complex environments or large aspect ratios, while also providing each

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TABLE V
QUANTITATIVE COMPARISONS OF OUR PROPOSED MODEL AND OTHER EVALUATED METHODS ON DIOR DATASET

Method	mAP	AL	AT	BF	BC	B	C	D	ESA	ETS	GC	GTF	HB	O	S	SD	ST	TC	TS	V	WM
Faster R-CNN [66]	66.9	81.1	76.9	89.8	80.5	45.1	81.7	63.4	80.9	63.2	73.3	78.3	43.3	58.4	68.5	**89.1**	59.0	81.1	60.2	47.8	79.3
Cascade R-CNN [67]	67.4	81.2	81.4	90.1	81.1	46.9	81.5	**68.1**	84.2	65.3	74.5	81.6	37.8	60.4	68.9	88.8	60.5	81.0	57.5	47.7	80.7
RetinaNet [12]	66.1	53.3	77.0	69.3	85.0	44.1	73.2	62.4	78.6	62.8	78.6	76.6	49.9	59.6	71.1	68.4	45.8	81.3	55.2	44.4	85.5
YOLO-v4 [9]	66.4	79.3	71.2	70.2	88.7	39.2	77.1	48.6	58.2	58.3	75.5	71.2	53.8	57.7	90.1	51.0	77.5	83.3	46.7	55.1	75.1
YOLO-v5 [68]	68.6	87.3	61.7	73.8	90.0	42.6	77.5	55.2	63.8	63.2	66.9	78.0	58.2	58.1	87.8	54.3	**79.3**	89.7	50.2	54.0	79.6
FCOS [20]	61.2	73.5	68.0	69.9	85.1	34.7	73.6	49.3	52.1	47.6	67.2	68.7	46.3	51.1	72.2	59.8	64.6	81.2	42.7	42.2	74.8
CornerNet [19]	65.9	58.8	**84.2**	72.0	80.8	46.4	75.3	64.3	81.6	**76.3**	79.5	79.5	26.1	60.6	37.6	70.7	45.2	84.0	57.1	43.0	75.9
CenterNet [18]	63.9	73.6	58.0	69.7	88.5	36.2	76.9	47.9	52.7	53.9	60.5	62.6	45.7	52.6	88.2	63.7	76.2	83.7	51.3	54.4	79.5
SRAF-Net [23]	69.7	88.4	76.5	**92.6**	87.9	35.8	**83.8**	58.6	86.8	66.8	76.4	**82.8**	16.2	58.0	59.4	80.9	55.6	90.6	52.0	53.2	**91.0**
MSFC-Net [69]	70.1	85.8	76.2	74.4	91.1	44.2	78.1	55.5	60.9	59.5	76.9	73.7	49.6	57.2	89.6	69.2	76.5	86.7	51.8	55.2	84.3
DETR [70]	68.6	39.6	74.0	65.2	80.7	26.5	75.2	66.8	70.5	52.5	74.2	62.1	27.4	47.0	8.5	46.0	14.5	64.4	54.3	14.4	55.6
DIAG-TR [71]	72.0	68.4	72.2	71.9	85.1	29.8	76.8	60.2	70.3	73.2	75.1	35.5	35.5	50.8	41.6	71.8	32.9	81.0	53.1	33.8	78.6
RAST-YOLO [72]	69.8	84.3	76.4	78.7	85.9	40.2	76.8	50.2	62.6	56.5	77.1	73.7	61.1	56.6	91.1	74.3	77.9	89.3	53.3	54.0	76.2
ASSD [73]	71.1	85.6	82.4	75.8	89.5	40.7	77.6	64.7	67.1	61.7	**80.8**	78.6	62.0	58.0	84.9	76.7	65.3	87.9	**62.4**	44.5	76.3
FSoD-Net [74]	71.8	88.9	66.9	86.8	90.2	45.5	79.6	48.2	**86.9**	75.5	67.0	77.3	53.6	59.7	78.3	69.9	75.0	**91.4**	52.3	52.0	90.6
TMAFNet [1]	72.9	**92.2**	77.7	75.0	**91.3**	**47.1**	78.6	53.6	67.1	66.2	78.5	76.3	64.9	61.4	90.5	72.2	75.4	90.7	62.1	55.2	83.2
Ours	**73.2**	90.9	72.8	79.3	89.7	44.7	81.4	59.3	66.0	62.7	73.8	79.2	**65.0**	**61.7**	**91.7**	78.6	75.8	90.7	60.0	**58.0**	83.1

TABLE VI
QUANTITATIVE COMPARISONS OF OUR PROPOSED MODEL AND OTHER EVALUATED METHODS ON HRRSD DATASET

Method	mAP	AL	S	ST	BD	TC	BC	GTF	HB	B	V	CR	TJ	PK
Faster R-CNN [4]	81.50	90.80	88.50	88.70	86.90	80.70	47.90	90.60	89.40	85.50	84.00	88.60	75.10	63.30
Cascade R-CNN [67]	89.10	98.00	91.40	95.60	91.30	92.60	72.20	98.20	93.90	89.60	95.50	92.30	79.70	68.40
YOLOv4 [9]	90.90	98.70	91.90	96.30	91.50	93.00	81.20	95.40	93.40	90.70	93.80	95.10	**84.90**	75.80
MGCN [75]	82.67	93.27	88.96	92.10	91.73	83.98	55.79	93.46	92.96	89.79	89.85	94.95	79.87	74.69
EGAT [31]	91.30	95.67	93.92	93.45	**94.52**	85.89	**86.63**	95.72	94.36	91.23	89.76	**95.86**	83.71	**86.22**
Ours	**92.03**	**99.79**	93.41	**98.05**	92.66	**96.12**	82.79	**99.06**	93.99	**93.82**	**96.60**	94.09	84.02	71.93

TABLE VII
QUANTITATIVE COMPARISONS OF OUR PROPOSED MODEL AND OTHER EVALUATED METHODS ON NWPU VHR-10 DATASET

Method	mAP	airplane	ship	ST	BD	TC	BC	GTF	harbor	bridge	vehicle
Faster R-CNN [66]	88.30	99.70	85.58	99.27	95.93	88.22	92.08	99.73	92.11	43.37	86.60
Cascade R-CNN [67]	92.19	99.54	88.53	95.98	94.46	94.02	88.21	97.16	91.45	82.30	90.25
YOLO-v4 [9]	90.77	99.40	92.63	**99.81**	95.27	98.30	75.22	94.17	93.12	71.95	87.80
YOLO-v5 [68]	92.3	99.50	94.50	76.40	99.00	90.50	97.20	99.50	**97.40**	76.00	92.70
YOLO-v7 [11]	93.31	99.74	92.71	97.69	97.87	93.97	91.83	95.63	89.51	83.75	90.47
YOLO-v8 [76]	93.95	99.99	94.37	96.78	98.12	94.36	90.77	97.25	91.33	85.7	91.02
FCOS [20]	92.14	99.99	85.21	96.94	97.75	95.80	80.34	99.67	95.04	81.82	88.92
CANet [77]	93.33	99.99	85.99	99.27	97.28	97.80	84.77	98.38	90.38	**89.16**	90.25
EGAT [78]	92.01	97.32	**96.72**	97.16	96.51	86.64	94.46	94.18	86.18	80.14	90.79
ABNet [30]	94.21	**100**	92.58	97.77	97.76	99.26	95.98	**99.86**	94.26	69.04	**95.62**
GLSANet [2]	94.57	99.90	95.80	97.12	**99.40**	98.75	86.12	99.46	97.51	84.83	86.76
Ours	**95.21**	99.88	94.23	95.47	98.43	**99.69**	**96.34**	99.33	96.88	84.57	87.25

subtask with essential information. By replacing the coupled detection head with the ECDD, we achieve an improved detection performance, with respective mAP values of 73.2%, 92.03%, 95.21%, and 94.30% on the DIOR, HRRSD, NWPU, and RSOD datasets, representing 0.6%, 0.59%, 0.45%, and 0.12%, respectively, mAP increase over the baseline network. Overall, ECDD exhibits competitive performance on RSI datasets.

D. Comparison Experiments and Analysis

In this section, we analyze the comparative experiments between attention-free global multiscale fusion network (AGMF-Net) and other state-of-the-art detectors on DIOR, HRRSD, NWPU VHR-10, and RSOD. The results are presented in Tables V–VIII.

1) Results on DIOR: Our method is compared with 16 advanced approaches, and the results are detailed in

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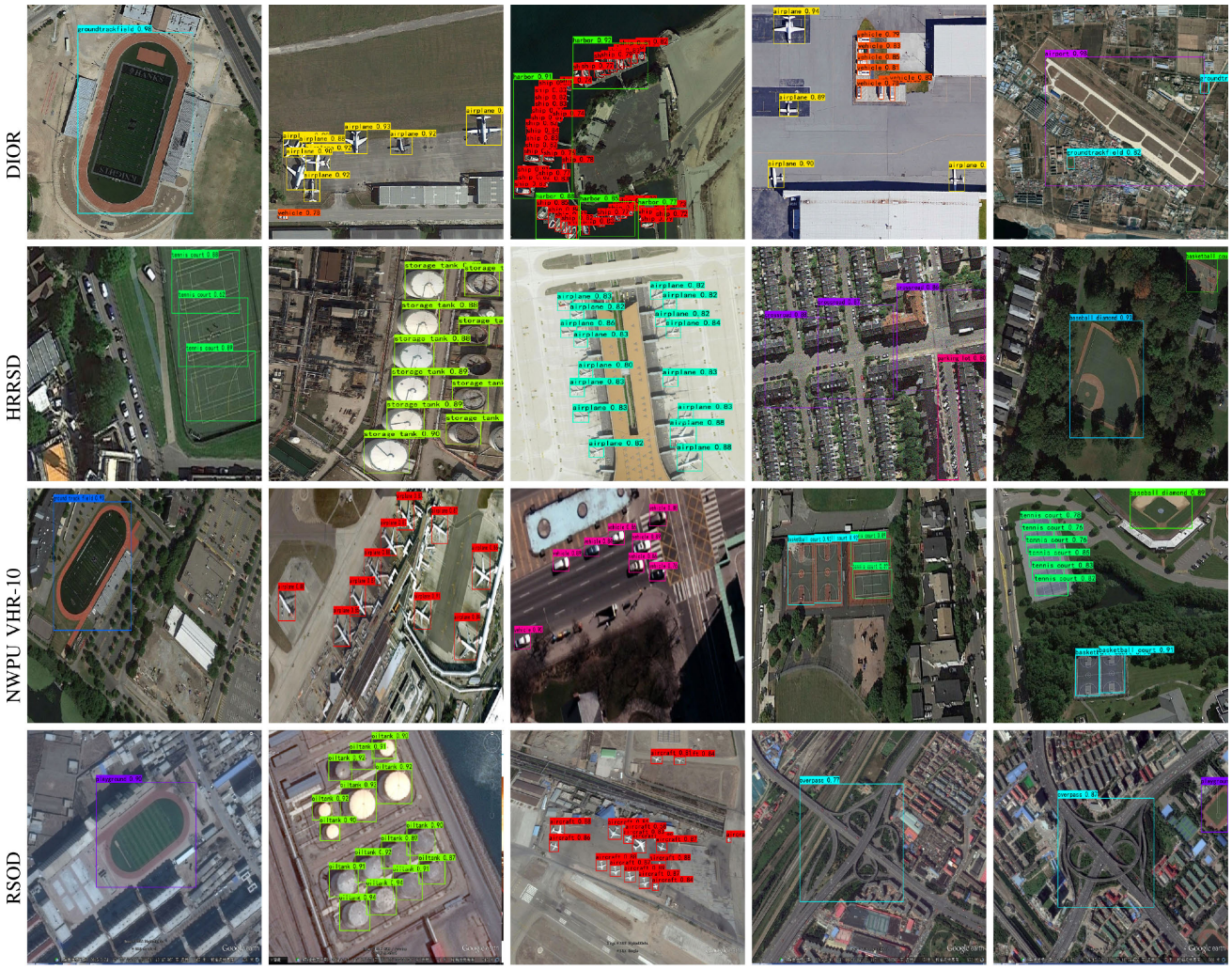


Fig. 6. Some representative detection results of AGMF-Net on the DIOR, HRRSD, NWPU VHR-10, and RSOD datasets. AGMF-Net excels in accurately detecting objects of various scales. (a) Large object, (b) medium object, (c) small object, (d) object in complex background, and (e) multiscale object within the same image. Since the images we selected contain different scale targets, zoomed-in view on the page gives a better visual effect.

TABLE VIII

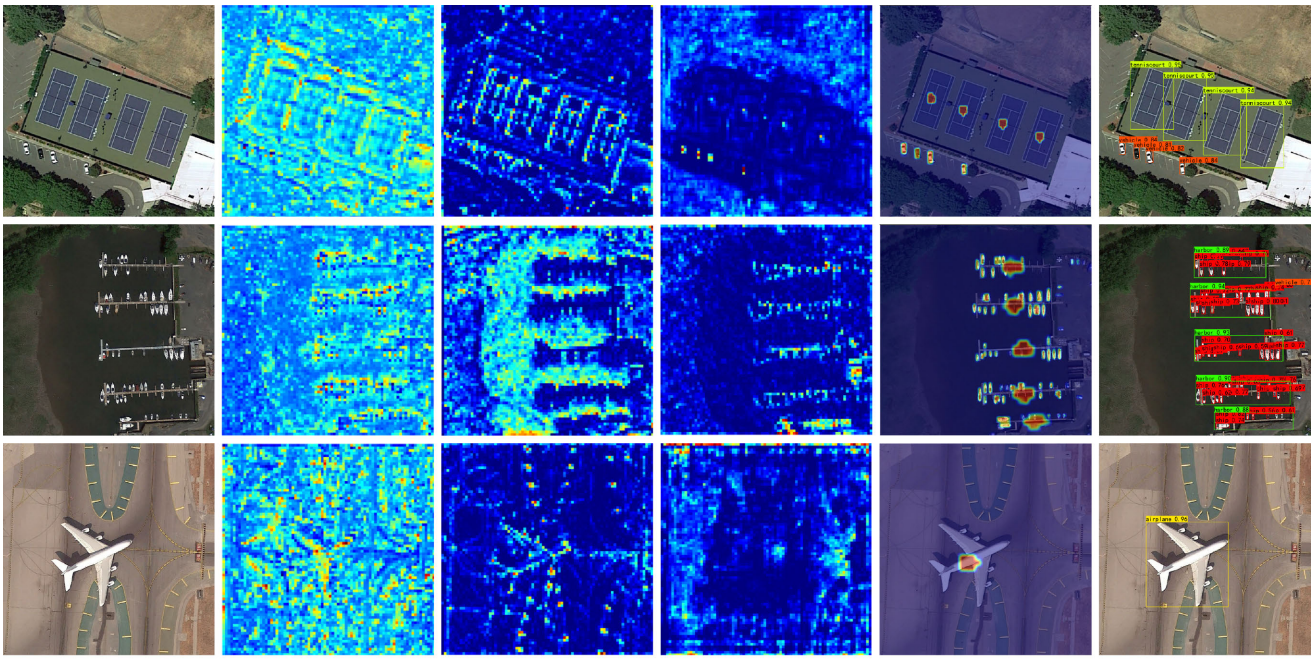
COMPARISON WITH STATE OF THE ARTS ON RSOD [79]

Method	mAP	Aircraft	Oil tank	Overpass	Playground
Faster RCNN [66]	84.47	70.84	90.19	78.74	98.09
Cascade R-CNN [67]	91.30	94.20	96.10	83.20	99.00
Deformable R-FCN [80]	87.92	71.87	90.35	**89.59**	99.88
soft-NMS [81]	86.60	76.10	90.30	81.30	98.80
Sig-NMS [82]	89.40	80.60	90.60	87.40	99.10
SSAFNet [83]	92.82	95.75	98.39	84.66	92.50
ATTS [84]	91.30	94.50	96.00	74.90	**99.90**
Ours	**94.30**	**96.02**	**99.02**	82.43	99.70

Table V. AGMF-Net achieves the highest mAP of 73.2%, outperforming all competitors. This highlights the effectiveness of our proposed method. In addition, our approach excels in three specific categories: HB (65.0%), overpass (61.7%), ship (91.7%), and vehicle (58.0%). Notably, the AP of small-scale vehicles is 2.8% higher than the second-place TMAFNet, demonstrating the improvement in small object detection achieved by our method.

2) Results on HRRSD: We evaluated AGMF-Net on HRRSD and compared it against the latest five methods. AGMF-Net stands out as the only method exceeding % mAP, as shown in Table VI. It achieves the best results in six out of 13 categories: 99.79% AP on an AL, 98.05% AP on the ST, 96.12% AP on the TC, 99.06% AP on a GTF, 93.82% AP on the bridge, and 96.60% AP on the vehicle. Our proposed method demonstrates effective object detection across various scales. Particularly in categories with numerous small objects, such as ALs and vehicles, and AGMF-Net outperforms all detectors. The AP on AL (99.79%) is 1.09% higher than the second-place YOLOv4, the AP on vehicle (96.60%) surpasses the second-place cascade R-CNN by 1.10%. Our network demonstrates higher training efficiency and superior target anchoring ability compared to two-stage networks and anchor-based networks. Furthermore, the visualization results in Fig. 6 effectively illustrate the accurate detection of large-scale and small-scale targets, as well as targets with diverse aspect ratios.

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(a) (b) (c) (d) (e) (f)

Fig. 7. Visualizing the feature extraction effects of GIEM, MFP, and MES. (a) Three original remote sensing images. (b) C3 feature extraction results from GIEM. (c) Feature pretreatment after MFP. (d) P3 feature map enhanced by MES (e) Combination effect of the heatmap and the original images. (f) Final detection results.

3) Results on NWPU VHR-10: In Table VII, we compare AGMF-Net's performance with other 11 state-of-the-art detectors on the NWPU VHR-10 dataset. AGMF-Net achieves the best average precisions (APs) in two out of ten categories, notably excelling in small-scale classes like TC (99.69%) and BC (96.34%). It secures the second place in the HB category (96.88%) at a smaller scale. However, it performs relatively poorly in detecting vehicles, attaining only 87.25% AP in this category. This could be due to the fact that the images containing vehicles are locally sharpened color infrared images, unlike the typical RSIs taken by satellites resulting in subpar performance in this class.

4) Results on RSOD: We compared our method with seven advanced approaches, as shown in Table III. AGMF-Net achieves the highest mAP of 94.30%, outperforming all competitors. This underscores the effectiveness of our proposed method. Moreover, it excels in two specific classes: aircraft (96.02%) and oil tank (99.02%). Notably, the AP of small-scale aircraft is 1.97% higher than that of the second-place ATTS, demonstrating our method's improvement in small object detection.

E. Detection Visualization Results

To further highlight the effectiveness of our proposed AGMF-Net, we visualize its prediction results on DIOR, HRRSD, NWPU VHR-10, and RSOD.

As shown in Fig. 6, it achieves excellent detection performance on DIOR. In Fig. 6(a) and (b), large-sized and medium-sized objects are easily and accurately detected. Almost all the small targets in Fig. 6(c) can be detected accurately. Fig. 6(d) shows the prediction results in complex backgrounds of the RSIs. In each image of Fig. 6(e), various objects of different sizes are successfully detected. We can note that although the RSIs of them contain large and complex backgrounds, our method still effectively detects the objects in them, especially the small-scale vehicles, ALs, and ships. The visualization results suggest that the superior performance of AGMF-Net in multiscale object detection on RSIs again.

F. Feature Extraction Visualization Results

To further illustrate the impact of GIEM and MES on feature extraction, we visualize the feature maps of small-scale TCs, ALs, and ships in Fig. 7. In Fig. 7(b), after applying GIEM, feature maps can capture object features but contain significant background features and noise (e.g., curves on the uptown area, the beach near HBs, and the AT ground), leading to interference in small object detection. In addition, the features of ships are jumbled together, making accurate localization challenging. In Fig. 7(c), after applying MFP, background and noise are notably suppressed, and features of TCs, ALs, and ships become richer. It is observed from Fig. 7(d) that the combination of MFP with MES architecture can make the output features clearer and more accurate. Despite dense ship distributions, MES significantly reduces feature confusion between neighboring ships, separating features of most ships effectively. In addition, ECDD can accurately differentiate target categories and label them accordingly. These visualizations demonstrate that GIEM and MES enhance features of remote sensing objects, especially small ones, while reducing the influence of background and noise. These modules contribute to improved object detection performance of ECDD.

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V. Conclusion

In this article, we introduce AGMF-Net, a novel one-stage object detection network designed to enhance multiscale object detection in RSIs, particularly for small-scale objects. AGMF-Net effectively aggregates global semantic information and local feature details through several key innovations. Initially, GIEM mines and enhances high-level semantic information in the deep feature map. Moreover, MES is designed to enhance the representation capacity of multi-scale targets while filtering out interference from complex backgrounds. In addition, ECDD is specifically designed for RSIs. It utilizes fused semantically strong features and high-resolution feature maps for both classification and localization tasks. Comparative experiments with consistent dataset settings across various detection methods demonstrate the superior performance of AGMF-Net, particularly in small object detection. AGMF-Net offers insights for addressing the challenges of complex backgrounds and small object detection in RSIs. We have observed that the detection performance of AGMF-Net on color infrared images is not satisfactory, and further investigation will be conducted in the future. Furthermore, the utilization of multimodal data to enhance efficiency in target detection tasks, such as leveraging visible light images and infrared images, has gained significant attention recently. We plan to explore this area in our future research endeavors.

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Tao Gao (Member, IEEE) received the B.Sc., M.Sc., and Ph.D. degrees from the School of Electronics and Information, Northwestern Polytechnical University, Xi'an, China, in 2002, 2016, and 2010, respectively.

He is currently a Professor with the School of Information Engineering, Chang'an University, Xi'an. His research interests include computer vision and intelligent transportation.



Ziqi Li received the B.Sc. degree from the School of Electronics and Information, Xi'an Technological University, Xi'an, China, in 2019. He is currently pursuing Ph.D. degree with the School of Information Engineering, Chang'an University, Xi'an.

His research interests include image processing and object detection.



Yuanbo Wen received the B.Sc. degree from the School of Information Engineering, Chang'an University, Xi'an, China, in 2021, where he is currently pursuing the Ph.D. degree.

With a specialized focus on computer vision and intelligent transportation, he is actively involved in cutting-edge research in these fields.



Ting Chen received the Ph.D. degree in information and communication engineering from Xidian University, Xi'an, China, in 2011.

She is currently an Assistant Professor with the School of Information Engineering, Chang'an University, Xi'an. Her research interests include image processing and wireless network.



Qianqian Niu received the B.Sc. degree in electronic information engineering from Chang'an University, Xi'an, China, in 2021, where he is currently pursuing the M.Sc. degree in information and communication engineering.

His research interests include remote sensing object detection (RSOD) and recognition.



Zixiang Liu received the B.Sc. degree from the School of Information Engineering, Chang'an University, Xi'an, China, in 2021, where he is currently pursuing the M.Sc. degree.

He is engaging in the research of computer vision and intelligent transportation.

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